Actual geometric p-torsion and j=0, in chars two and three

Result. Every nonsingular general Weierstrass model over every algebraically closed field of characteristic p=2 or3: actual subgroup killed by p has count1 if j=0, otherwise p; the actual induced ZModp module has rank0 if j=0, otherwise1. Equivalently, j=0 iff every actual killed-p point is the identity. Every algebraically closed scalar extension has the count determined by the base Mathlib j-invariant. Actual IsElliptic means unit discriminant, equivalent to nonzero discriminant over a field. No rank, basis, normal-form, trace or pairing premise. [pcc:unverified]

Same F3 curve: $y^2=x^3+2x^2+1$, actual $\Delta=1$ / $j=1$ / rational point count=5

Actual proof bridges. In characteristic three, the literal division polynomial evaluates to $b2*x^3+b8$. If $b2=0$, the actual discriminant/invariant relation forces $b8\neq0$, so only the identity is killed by three. If $b2\neq0$, construct an actual cube root and ordinate, hence a nonzero nonsingular point killed by three; the previous actual one-or-three classification forces three. Combine Mathlib's existing $j=0$ iff $b2=0$. In characteristic two reuse the earlier actual subgroup classification and official $j=0$ iff $a1=0$. Construct the actual two-torsion module; finite counts determine module ranks. Official map_j and injective field maps prove scalar-extension criteria. [pcc:unverified]

Lean negative control. The actual F3 curve above has five rational points, so its rational killed-three subgroup is trivial, despite $j=1$. Its geometric killed-three subgroup has three points. Kernel-checked point enumeration and an actual group-order argument establish these facts. Lean negates the concrete iff and the universal statement obtained by dropping algebraic closure. This tests a deleted hypothesis and does not refute the qualified geometric criterion. [pcc:unverified]

Primary source and qualifications. Milne, Elliptic Curves, selected Theorem3.12/context/Remark3.16, printedpp98–99, and notes p75; PDFpages104/80 rendered and inspected. The remark describes algebraically closed p-power torsion ranks, while p75 describes endomorphism-rank terminology. No equivalence is assumed here; only p itself in characteristics2/3 is handled. Primary book URL/hash retained locally, no raw book republished or full-book reading claimed. Prior actual torsion proofs and official Mathlib coefficient identities credited. KB/Conductor unavailable; bounded fallback recorded. [pcc:unverified]

Scope kept open. The original prime-field node already requires $p\geq5$, so characteristic-three examples cannot refute it. Existing conditional trace-zero arithmetic is credited. The FST trace-based definition, endomorphism-rank equivalence, Hasse/trace geometry, general embedding degree, other-prime geometry/full all-k $p<500$ census and remaining fourteen-item obligations remain open. [pcc:unverified]

Validation. Whole gate30/0failures/0trusted native exceptions; complete declaration binding/recursive source-config pins. All62attempt-provenance files retained losslessly in24objects; five gate records and failed source/config/log/exit snapshots. Pinned Lean4.35.0/Mathlib56a00585…, only standard foundations. Exact local PCC replay is separate from signed standing. No novelty, qualified-source refutation, corpus promotion, native experiment, paid compute, upload, holes/custom axioms/native evaluator/raised limits/new global flags/waivers. [pcc:unverified]